Zadanie 6 [3 pkt] – Typ niestandardowy

Oblicz $\left[3^{\frac{1}{2}} - 0{,}25(3^{1{,}5} - 3^{-0{,}5})\right] : 6^{-0{,}5}$

W tym zadaniu warto skorzystać ze wzorów:

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} \qquad a^{-n} = \frac{1}{a^n}$$

$$\left[\sqrt{3} - \frac{1}{4}\left(3^{\frac{3}{2}} - 3^{-\frac{1}{2}}\right)\right] : 6^{-\frac{1}{2}} = \left[\sqrt{3} - \frac{1}{4}\left(3\sqrt{3} - \frac{1}{\sqrt{3}}\right)\right] : \frac{1}{\sqrt{6}} =$$

usuwamy niewymierności

$$= \left[\sqrt{3} - \frac{1}{4}\left(3\sqrt{3} - \frac{\sqrt{3}}{3}\right)\right] : \frac{\sqrt{6}}{6} = \left[\sqrt{3} - \frac{1}{4}\cdot\left(\frac{9\sqrt{3}}{3} - \frac{\sqrt{3}}{3}\right)\right] : \frac{\sqrt{6}}{6} =$$

$$= \left[\sqrt{3} - \frac{1}{4}\cdot\frac{8\sqrt{3}}{3}\right] : \frac{\sqrt{6}}{6} = \left[\sqrt{3} - \frac{2\sqrt{3}}{3}\right] : \frac{\sqrt{6}}{6} = \left[\frac{3\sqrt{3}}{3} - \frac{2\sqrt{3}}{3}\right] : \frac{\sqrt{6}}{6} =$$

$$= \frac{1}{3}\sqrt{3} \cdot \frac{6}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{1}{3}\sqrt{3} \cdot \frac{\cancel{6}\sqrt{6}}{\cancel{6}} = \frac{1}{3}\sqrt{3}\cdot\sqrt{6} = \frac{1}{3}\cdot\sqrt{3\cdot 6} =$$

ponownie usuwamy niewymierność

$$= \frac{1}{3}\cdot\sqrt{18} = \frac{1}{3}\cdot\sqrt{9\cdot 2} = \frac{1}{3}\cdot 3\sqrt{2} =$$

$$= \sqrt{2}$$

Odp: $\sqrt{2}$.